Project 1 - Northwest Architecture Studio Ltd.

Introduction

The company needs new PCs for distinct types of staff. It does not want to buy the same PC for every employee because different roles have different computing requirements. Some staff only need basic office computers, while others need more powerful systems for CAD, 3D modelling, and visualisation.

We will:

1. describe each role, including what type of tasks they do and what type of PC they require to complete these tasks efficiently.

2. make a full descriptive workstation build for each of the four roles, we will detail every component with specification, cost, and justification.

3. create a final cost table. We will enter the budget for each build, how much the actual build costs, and any leftover funds with reasoning.

4. create an SQL database with linked tables, allowing the user to query the database for whatever information they require. We will also explain how the database has strong integrity.

5. talk about our ethics/professionalism when spending the company's funds.

We will not include operating system licences, office software, CAD software, antivirus subscriptions, and other software licences towards the budget as the company's internal IT team manages these.

Role Description

Before we create the PC builds, we must find out what the typical work for each role is. For example, an administrative assistant would only need basic components for tasks involving emails, documents and meetings whereas a 3D Visualisation Specialist would require high-end components to create large models and designs. This allows us to narrow down our search, making it easier to find suitable hardware for the

specific roles - we wouldn't want to waste funds on an expensive high-end GPU for a role that only requires the basic functions to complete their tasks. Stops fund waste and is good professional practice.

Role	Typical work	Hardware Requirements Justification
Administrative Assistant - 3 workstations	Email, documents, spreadsheets, web systems, online meetings, and basic customer records.	They work with emails, documents, spreadsheets, web systems, online meetings, and basic customer records. They do not need a high-spec machine, but a basic yet reliable workstation that allows them to smoothly/efficiently do their job.
Project Manager - 2 workstations	Reports, meetings, project planning, reviewing drawings, dashboards, and client communication.	Project Managers write reports, run meetings, plan projects, review drawings, use dashboards and communicate with clients. They multitask heavily, with many windows open and frequent screen sharing. They benefit most from screen space and good communication kit rather than raw processing power, and they do not run CAD or rendering software.
Architect / CAD Designer - 5 workstations	CAD drawings, 2D drafting, 3D modelling, large design files, multiple monitors, and design reviews.	Architects and CAD Designers produce 2D drawings, 3D models and large design files, often with several windows or multiple monitors and frequent design reviews. Their software needs fast CPU cores, a dedicated GPU, plenty of RAM, fast storage and generous screen space, so their workstations need to be stronger than the office builds but do not need the extreme rendering power of the 3D Visualisation role.
3D Visualisation Specialist - 2 workstations	Rendering, 3D visualisation, large models, high-resolution textures,	3D visualization specialist workstation required modern GPU to accelerate rendering speed, gigabyte RTX GeForce 5060 MAX OC 8G is a good choice for limited budget. AMD Ryzen 9 9950X3D has 16 cores / 32 threads which process multiple tasks faster, support PCIe 5.0 and DDR5 memory

	media output, and client visual presentations.	support. A 3D Visualisation required large memory, higher resolution textures, 64GB RAM allows to perform simultaneously.

Hardware Specification + BuildCores Screenshot + Peripherals + Cost Table + Component Justification

In this section, we will detail the exact components for each role's chosen PC workstation including specifications, cost, and justification. We will NOT include operating system licences, office software, CAD software, antivirus subscriptions, and other software licences towards the budget as the company's internal IT team manages these.

The details will be shown in a table format for easy comprehension.

Administrative Assistant:

This build only requires the basic and essential hardware. It does not need a dedicated GPU, role-specific hardware, or high-end PC specifications.

Hardware	Specification	Cost + last checked	Justification
CPU	Intel Core i3-12100 4 Cores / 8 Threads 3.3 GHz LGA1700 Socket	£120 28/09	4 Cores / 8 Threads allows the user to multi-task comfortably, for example if they had to access documents whilst being in an online video call. This comes with an integrated GPU (Intel UHD Graphics 730), so no further costs will need to be spent on a dedicated GPU - a basic workstation has no need for it. Additionally, it includes a reliable CPU cooler (Intel Laminar RM1 Boxed Cooler), saving even more funds that would otherwise be spent on a separate cooler.

			https://business.currys.co.uk/catalogue/computing/components-upgrades/processors/intel-core-i3-12100-3-3-ghz-processor-box/P282453P
Motherboard	**MSI PRO H610M-E DDR4 (Micro-ATX)** Intel H610 Chipset LGA1700 Socket 2x DIMM slots, up to 64GB at 3200MHz 1x PCIe 4.0 x16 slot, 1x M.2 NVMe slot, 4x SATA 6Gbps ports HDMI 1.4, VGA, USB3.2 Gen 1 Realtek Gigabit Ethernet LAN	**£60** 28/09	Intel H610 chipsets provides the user with all the capabilities needed for administrative tasks. This includes an M.2 slot for fast NVMe SSD storage, USB 3.2 ports for high-speed file transfers, and Gigabit Ethernet for reliable connection. Additionally, it holds HDMI and VGA video outputs directly on the motherboard's rear panel which allows dual-monitor support. This is in case the business wants to implement another monitor to the workstation in the future, for whatever reason. A high-quality motherboard is not needed for someone who will only be using basic functions. https://www.awd-it.co.uk/msi-pro-h610m-e-ddr4-micro-atx-motherboard.html
RAM	**CORSAIR Vengeance LPX Black** 16GB (2x8GB) DDR4 3200MHz CL16 Dual Channel	**£145** 29/09	16GB provides enough memory to efficiently run web browsers with multiple tabs open. It is also enough for the user to multitask without slowing down the system e.g. accessing a spreadsheet whilst being in an online meeting. Having two 8GB RAM sticks instead of one 16GB RAM stick doubles the memory bandwidth, this is important because it boosts the overall performance of the system. Some benefits include improved responsiveness and increased CPU efficiency. https://www.scan.co.uk/products/16gb-2x8gb-corsair-ddr4-vengeance-lpx-black-pc4-25600-3200-non-ecc-unbuffered-cas-16-20-20-38-xmp-20
Storage	**Kingston NV3 M.2 2280 - SNV3S/1000G** 1TB NVMe SSD Read 6000MB/s, Write 4000MB/s PCIe 4.0	**£150** 29/09	Uses PCIe 4.0 technology to deliver high read/write speeds. This ensures fast boot times, application launches, and file sharing – highly beneficial, will most definitely boost productivity as the user won't have to waste time waiting for files and applications to load/open. 1TB SSD provides a very large amount of space for the user to store documents, applications, web systems, spreadsheets, etc. They will most likely never have the issue of running out of space. https://www.awd-it.co.uk/kingston-nv3-pcie-4-0-nvme-ssd-1tb-m-2-2280-snv3s-1000g-read-6000mb-s-write-4000mb-s.html
GPU	**Integrated Intel UHD Graphics 730**	**£0**	The CPU (Intel Core i3-12100) includes this built-in integrated graphics card, which eliminates the need to spend extra funds on an unnecessary dedicated graphics card. This GPU is more than enough for an Administrative Assistant's daily workload, ensuring smooth performance in online video calls and when working with large documents and systems.
PSU	**MSI MAG A550BN 550W 80 PLUS Bronze Power Supply**	**£45** 29/09	550W of continuous output at 80 PLUS Bronze efficiency. This gives the system even more power headroom, whilst keeping operating temperatures and fan noise minimal under typical office workloads.

			MSI is a highly regarded brand in the tech industry, they have built this PSU with high-level safety features to prevent power surges/fluctuations that would otherwise cause great damage to the workstation and possibly even data. https://www.awd-it.co.uk/msi-mag-a550bn-550w-80-bronze-rated-atx-power-supply-unit.html
Case/cooling	**Fractal Design Core 1100 Black Micro-ATX Mini Tower Case - Black + Intel Laminar RM1 Boxed Cooler**	**£40 + £0** 29/09	Compact mini-tower layout. Fits perfectly into office environments, clean, and covers all the components inside. Doesn't take up excessive space on the desk. The Fractal comes with a pre-installed front intake fan and dust filter for adequate airflow. Combined with the CPU's included Intel Laminar RM1 CPU boxed cooler, it easily keeps the Intel Core i3-12100 running cool and quiet without spending extra on over-the-top coolers. https://www.scan.co.uk/products/fractal-design-core-1100-micro-atx-mini-tower-case-w-o-psu-(std-atx)
Monitor(s)	**PHILIPS 24E1N1200A IPS Monitor – Black** Full HD 1920 x 1080p 24" 120Hz refresh rate 1ms response time Built-in speakers Wall mount compatibility Input: HDMI / VGA / DisplayPort	**£75** 29/09	IPS panel provides wide viewing angles and colour accuracy without distortion, reducing eye strain during work hours and preventing the user's eyes from deteriorating over time. This is especially important for an Administrative Assistant because most of their hours will be spent in front of the screen – editing files and spreadsheets, attending online video calls, accessing web systems, etc. Full HD 1920 × 1080p resolution delivers sharp, clear visuals, whilst the 120Hz refresh rate ensures smooth, fluid motion when scrolling and reading small text. It also comes with built-in speakers, so extra costs won't have to be spent on a microphone/headset which can also clutter the user's desk. https://www.currys.co.uk/products/philips-24e1n1200a-full-hd-24-ips-monitor-black-10280649.html
Keyboard/mouse	**Logitech Combo MK270 Combo Wireless Desktop Keyboard and Mouse USB**	**£25** 29/09	Having a wireless KB/M eliminates cable clutter on the desk, creating a clean and neat environment for the user to work in. The keyboard is integrated with a number pad, which can be useful when working with spreadsheets and other data entry responsibilities. To connect both the mouse and keyboard to the PC, it only requires one USB slot. This is beneficial as it creates space for other peripherals that may need a USB slot to connect. This product is also known for its durability and long battery life (up to 36 months for keyboard, 12 months for mouse), reducing operational upkeep costs. https://www.scan.co.uk/products/logitech-mk270-wireless-desktop-compact-spill-resistant-multimedia-keyboard-and-optical-mouse-1000dp
Webcam	**Logitech Brio 100 Full HD Webcam – Graphite** 1080p Auto-Light Balance Built-in Privacy Shutter Integrated Microphone	**£40** 29/09	Sharp 1080p Full HD video for online video calls. Auto-light balance brightens video if office lighting is dim or too bright, and the built-in privacy shutter allows quick physical security that covers the camera when off-call. https://www.currys.co.uk/products/logitech-brio-100-full-hd-webcam-graphite-10254678.html

Headset	**Logitech H390 Stereo MIC USB Black** Noise-Cancelling In-line Controls Leatherette Ear Cushions USB-A Plug-and-Play	**£25** 29/09	Noise-cancelling microphone used to isolate the user's voice, especially useful in loud background office environments. Leatherette ear cushions will ensure the user's comfort, specifically for those who are dealing with customer calls or video meetings for a long period of time. https://www.scan.co.uk/products/logitech-h390-stereo-headset-with-microphone-black-wired-noise-cancelling-usb-pc-mac
Network/cables/adapters	**TP-Link Archer T2U Plus** USB 2.0 Dual Band Wi-Fi 5 802.11ac 2.4 GHz: 200 Mbps, 5 GHz: 433 Mbps External 5dBi High-Gain Adjustable Antenna	**£15** 29/09	High-Gain Adjustable Antenna offers a significantly boosted wireless connection, increased range, and signal stability. Dual Band speeds up to 433Mbps (5GHz band) allow fast file transfers, smooth video conferencing etc. This product can be directly attached to the motherboard for high-speed ethernet connections and Dual Band capability. https://business.currys.co.uk/catalogue/computing/servers-networking/networking/router-accessories/tp-link-archer-t2u-plus-network-adapter-usb-2-0/P301931P
Role-specific hardware peripheral	N/A	**£0**	This role does not require any specific hardware peripherals needed for their work. They will rely on standard office tools like email, spreadsheets, video calls etc. All necessary peripherals have already been provided and justified (keyboard, mouse, webcam, headset, monitor)
Total hardware workstation cost		**£740**	
Role hardware budget		**£950**	
Under/over hardware budget		**Under £210**	

£740 for one complete workstation. For 3 workstations, it will cost £2220. The total role budget is £2850, so there will be £630 leftover. OS, application, antivirus, CAD, cloud, and subscription costs have been excluded as required.

Project Manager:

Cost table and justification

Hardware	Chosen item/specification	Cost	Justification
CPU	Intel Core i5 14400 (10 cores / 16 threads, integrated UHD 730 graphics)	£159.99	Gives plenty of multitasking headroom for large PDFs, dashboards, planning tools and Teams calls running together. The built-in graphics can drive two monitors, so no separate GPU is needed for reviewing drawings, which saves money and power.
Motherboard	MSI B760 PRO M-P DDR4 Micro ATX	£89.99	A B760 board supports the CPU and the cheaper DDR4 memory, and has video outputs for the integrated graphics. The micro-ATX size keeps the case and cost down.
RAM	Corsair Vengeance LPX 16GB (2x8GB) DDR4 3200 CL16 Black	£165.00	16GB is enough for reports, meetings, dashboards and many browser tabs. DDR4 is used because DDR5 prices are currently very high. If project files grow, the memory can be upgraded to 32GB later by fitting larger modules.
Storage	Kingston NV3 PCIe 4.0 NVMe SSD 2280 1TB	£128.99	Holds local copies of project files, drawings and reports with room to grow, and gives fast start-up and quick opening of large PDFs.
GPU	None - integrated graphics	£0.00	Integrated graphics is enough for two monitors and drawing review. A dedicated GPU would exceed what this role needs.
PSU	Corsair RM650e (2025) Black 650W Fully Modular	£73.99	650W gives ample headroom for this low-power system and leaves room to add a graphics card later if the role changes. Gold-rated efficiency reduces running costs, and modular cables keep the build tidy and improve airflow.
Case/cooling	Corsair iCUE 4000D RGB AIRFLOW Mid-Tower ATX Case - True White (£129.00); Intel boxed cooler supplied with the CPU (£0)	£129.00	Good airflow with pre-installed fans keeps the machine cool and quiet in a shared office. The boxed cooler is adequate for this CPU, so no extra cooler is bought.

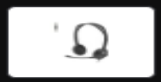
Monitor(s)	2 x KOORUI Curved 27-inch QHD Monitor, 180Hz, HDR10, VESA Adaptive-Sync (£183.81 each)	£367.62	The brief gives this role a stronger display setup. Two QHD screens give room for drawings, dashboards and reports on one screen with email or a video call on the other.
Keyboard	Logitech K120 UK Layout Wired Standard Keyboard	£10.77	A reliable full-size keyboard for report writing, at a low price that avoids over-spending.
Mouse	Logitech M705 Wireless Laser Black Mouse	£46.30	Comfortable and accurate for long working days, with a long battery life.
Webcam	ToLuLu 1080P Webcam with Microphone	£17.72	Project managers spend much of their time on client and team calls, so a 1080p camera with a built-in microphone is needed.
Headphones	Logitech H111 Closed-Back Headphones	£8.31	Closed-back headphones reduce background noise during online meetings.
Network/Cables/Adapters	DisplayPort cable (£10) + HDMI cable (£10) + 2m Cat6 Ethernet cable (£5)	£25.00	Needed to connect two monitors to the motherboard's video outputs and to give a stable wired network connection.
Total hardware workstation cost		£1,222.68	
Role hardware budget (per workstation)		£1,300.00	
Under/over hardware budget		Under £77.32	
Role quantity		2	

Total role cost		£2,445.36	2 x £1,222.68, against a role budget of £2,600.00 (under £154.64)

PC BUILD

Case		Corsair iCUE 4000D RGB AIRFLOW Mid-Tower ATX Case - True White	In stock	£129.00	OCUK	Buy
CPU		Intel Core i5 14400	In stock	£159.99	B&H	Buy
Motherboard		MSI B760 PRO M-P DDR4 Micro ATX	In stock	£89.99	Overclockers UK	Buy
GPU		+ Add GPU				
RAM		Corsair Vengeance LPX 16GB (2×8GB) DDR4 3200 CL16 Black	In stock	£165.00	amazon	Buy
		+ Add Additional				
CPU Cooler		Sponsored: MEG E15 – Stunning 110° View				+ Add CPU Cooler
Storage		Kingston NV3 PCIe 4.0 NVMe SSD 2280 1TB	In stock	£128.99	CCL	Buy
		+ Add Additional				
Power Supply		Corsair RM650e (2025) Black 650W Fully Modular	In stock	£73.99	CCL	Buy

SETUP

Monitor		KOORUI Curved 27-inch QHD Monitor, 180Hz, HDR10, VESA Adaptive-Sync	In stock	2 × £183.81	amazon	Buy
Keyboard		Logitech K120 UK Layout Wired Standard Keyboard	In stock	£10.77	B&H	Buy
Mouse		Logitech M705 Wireless Laser Black Mouse	In stock	£46.30	amazon	Buy
Webcam		ToLuLu 1080P Webcam with Microphone, HD Webcam Web Camera with Tripod Stand, Widescreen USB Computer Camera, Streaming Mic Web cam for Onlin...	In stock	£17.72	amazon	Buy
Headphones		Logitech H111 Closed-Back Headphones	In stock	£8.31	B&H	Buy

+ Add Additional



VIDEO

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Stouchi DisplayPort Cable 2.1 1.2M Grey, 80Gbps, 16K 60Hz, 8K, 4K 240Hz

Gaming monitor cable supports 16K@60Hz, 8K@120Hz and 4K@240Hz; HDR, HDCP 2.2, DSC 1.2a, FreeSync/G-Sync

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4.6 ★★★★☆ (1,647)

3 sustainability features

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1 sustainability feature

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Cat 6 Ethernet Cable 0.5m 1m 2m 3m 5m 10m 15m 20m 30m 40m 50m 65m 75m 85m 100m 150m 200m 250m 300m, High Speed 10Gbps Internet Cable, UTP 26AWG Black RJ45 LAN Patch Cables

Brand: JARNHNG

4.4 ★★★★½ (1,816)

50+ bought in past month

£4.99

Architect / CAD Designer:

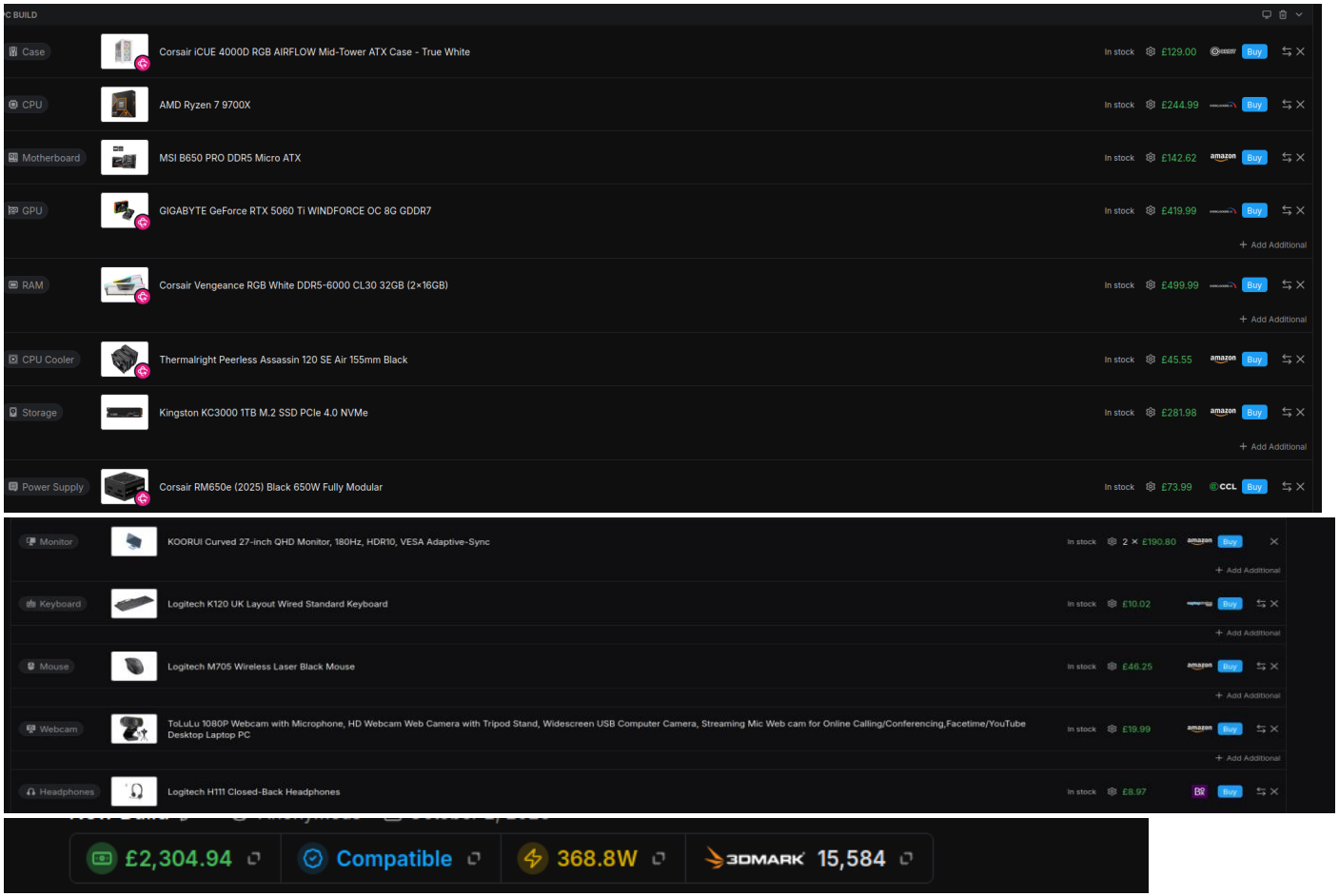
Cost Table and Justification

Hardware	Chosen item/specification	Cost	Justification
CPU	AMD Ryzen 7 9700X	£244.99	CAD and BIM software depends heavily on single-core speed, with extra cores helping exports and regeneration. The 9700X has very strong per-core performance at a 65W TDP, and the AM5 socket keeps a CPU upgrade path open.

Motherboard	MSI B650 PRO DDR5 Micro ATX	£142.62	A mid-range B650 board provides everything this role needs without paying for premium features. The AM5 socket and DDR5 support match the CPU and memory, and the micro-ATX size keeps the case and cost down.
RAM	Corsair Vengeance RGB White DDR5-6000 CL30 32GB (2x16GB)	£499.99	Large drawings and BIM models with several applications open at once need 32GB. DDR5 is required by the AM5 platform. Memory prices are currently high, and this cost reflects that. Two matched modules run in dual-channel mode.
Storage	Kingston KC3000 1TB M.2 SSD PCIe 4.0 NVMe	£281.98	Large model files open and save quickly. A drive with a DRAM cache holds up better under sustained writes than a budget drive, and 1TB leaves room for local project files.
GPU	GIGABYTE GeForce RTX 5060 Ti WINDFORCE OC 8G GDDR7	£419.99	A dedicated GPU is needed for 3D modelling and driving two monitors. 8GB of video memory is enough for typical architectural CAD and BIM models at a moderate price. A professional ISV-certified card would cost considerably more, so a GeForce card was chosen for value.
PSU	Corsair RM650e (2025) Black 650W Fully Modular	£73.99	650W gives comfortable headroom for the GPU and CPU. Gold-rated efficiency reduces running costs, and modular cables keep the build tidy and improve airflow.
Case/cooling	Corsair iCUE 4000D RGB AIRFLOW Mid-Tower ATX Case - True White (£129.00) +	£174.55	The 9700X is sold without a cooler, so one is required. A tower air cooler is quiet and low-maintenance, and the mesh-front case with pre-installed fans gives the GPU good airflow during long working sessions.

	Thermalright Peerless Assassin 120 SE Air 155mm Black (£45.55)		
Monitor(s)	2 x KOORUI Curved 27-inch QHD Monitor, 180Hz, HDR10, VESA Adaptive-Sync (£190.80 each)	£381.60	Dual monitors let architects keep drawings on one screen and tools, references or email on the other. QHD gives more workspace than 1080p at a moderate price, and VESA mounting allows adjustable arms.
Keyboard	Logitech K120 UK Layout Wired Standard Keyboard	£10.02	A full-size keyboard with a numeric keypad speeds up CAD data entry, and the low price avoids over-spending on a basic input device.
Mouse	Logitech M705 Wireless Laser Black Mouse	£46.25	Accurate tracking and extra buttons for CAD shortcuts, with long battery life.
Webcam	ToLuLu 1080p Webcam with Microphone	£19.99	Design reviews with clients and colleagues take place online, so a 1080p camera with a built-in microphone is needed.
Headphones	Logitech H111 Closed-Back Headphones	£8.97	Closed-back headphones reduce background noise during online meetings and when working in a shared studio.

Network/Cables/Adapters	2 x DisplayPort 1.4 cables (£10 each) + 2m Cat6 Ethernet cable (£5)	£25.00	Needed to connect two monitors and to give a fast, stable wired network connection for transferring large files.
Role-specific peripheral	3Dconnexion SpaceMouse Compact (3D navigation device)	£110.00	Speeds up orbiting and panning in 3D models and reduces mouse strain, so it is justified for a CAD role.
Total hardware workstation cost		**£2,439.94**	
Role hardware budget (per workstation)		**£2,500.00**	
Under/over hardware budget		**Under £60.06**	
Role quantity		**5**	
Total role cost		**£12,199.70**	5 x £2,439.94, against a role budget of £12,500.00 (under £300.30)


C BUILD
Case
Corsair iCUE 4000D RGB AIRFLOW Mid-Tower ATX Case - True White
In stock £129.00 Buy
CPU
AMD Ryzen 7 9700X
In stock £244.99 Buy
Motherboard
MSI B650 PRO DDR5 Micro ATX
In stock £142.62 Buy
GPU
GIGABYTE GeForce RTX 5060 Ti WINDFORCE OC 8G GDDR7
In stock £419.99 Buy
Add Additional
RAM
Corsair Vengeance RGB White DDR5-6000 CL30 32GB (2×16GB)
In stock £499.99 Buy
Add Additional
CPU Cooler
Thermalright Peerless Assassin 120 SE Air 155mm Black
In stock £45.55 Buy
Storage
Kingston KC3000 1TB M.2 SSD PCIe 4.0 NVMe
In stock £281.98 Buy
Add Additional
Power Supply
Corsair RM650e (2025) Black 650W Fully Modular
In stock £73.99 Buy
Monitor
KOORUI Curved 27-inch QHD Monitor, 180Hz, HDR10, VESA Adaptive-Sync
In stock 2 × £190.80 Buy
Add Additional
Keyboard
Logitech K120 UK Layout Wired Standard Keyboard
In stock £10.02 Buy
Add Additional
Mouse
Logitech M705 Wireless Laser Black Mouse
In stock £46.25 Buy
Add Additional
Webcam
ToLuLu 1080P Webcam with Microphone, HD Webcam Web Camera with Tripod Stand, Widescreen USB Computer Camera, Streaming Mic Web cam for Online Calling/Conferencing,Facetime/YouTube Desktop Laptop PC
In stock £19.99 Buy
Add Additional
Headphones
Logitech H111 Closed-Back Headphones
In stock £8.97 Buy
£2,304.94
Compatible
368.8W
3DMARK 15,584

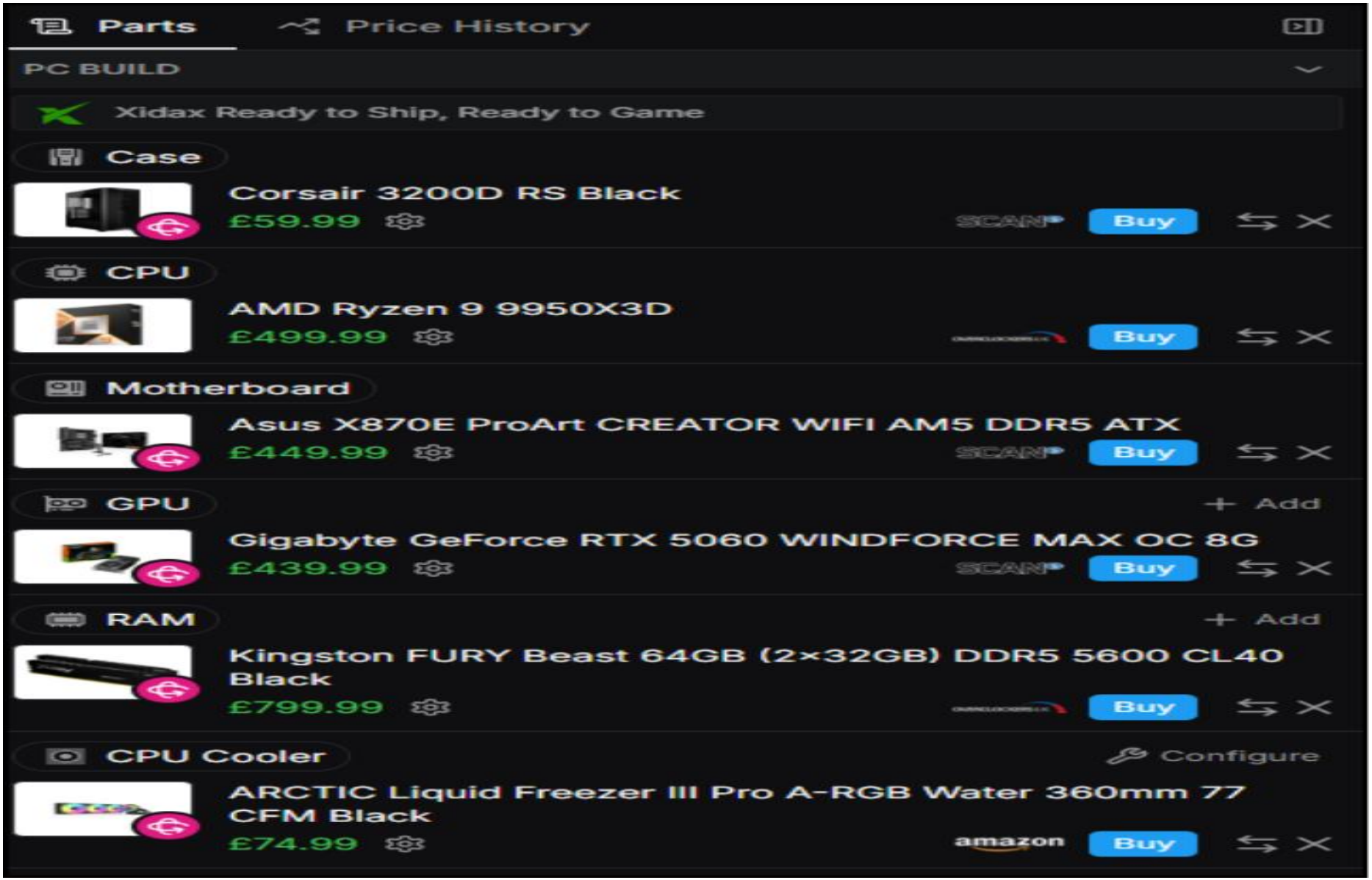
3D Visualisation Specialist:

Hardware	Quantity	Specification	Cost	Component link	Justification
Case (Corsair 3200D RS Black)	2	Support motherboard size ATX, Micro ATX, Mini ATX. Has 7 expansion slots.	£59.99	Corsair 3200D RS Black(O CFFICESTATION)	This case is compatible with my choice of ATX motherboard, case supports three Motherboard size, ATX, Micro-ATX, Mini-ATX
CPU (AMD Ryzen 9 9950X3D)	2	AM5 socket with a total of 16 Cores, 32 threads.	£499.99	AMD Ryzen 9 9950X3D	This CPU has 16 cores which makes work faster.
Motherboard (Asus X870E Por Art creator Wi-Fi AM5 DDR5 ATX)	2	Max Memory 256 GB , one temperature header, 8-pin CPU power	£449.99	Asus X870E ProArt CREATOR WIFI AM5 DDR5 ATX	This Motherboard is an ATX form factor, DDR5, because 3D requires large memory.
GPU (Gigabyte GeForce RTX 5060 wind force Max OC 8G)	2	Core Base clock is 2280 MHz,	£439.99	Gigabyte GeForce RTX 5060 WINDFORCE MAX OC 8G	The memory type is GDDR7 which supports high resolution needed for 3D visualization.
RAM (Kingston FURY Beast 64 GP (2*32 GP) DDR5 5600 CL40 Black)	2	DDR5 type, 5600 MHz speed, 288 pin DIMM, 32 GB	£799.99	Kingston FURY Beast 64GB (2x32GB) DDR5 5600 CL40 Black	64 GB RAM allows to perform complex task without slowing down with its capacity to supports 3D visualisation software's that use large memory space.

CPU Cooler (ARCTIC Liquid Freezer III Pro A- RBB Water 360mm 77 CFM Black)	2	Type: water cooling, Fan size: 120 mm. Support AM4, AM5, LGA 1700, LGA 1851.	£74.99	ARCTIC Liquid Freezer III Pro A-RGB Water 360mm 77 CFM Black	Rendering and multitasking are required higher cooling capacity (ARCTIC Liquid Freezer) is designed for high-performance CPUs, the radiator size for this cooler is 360 mm which offers large surface. The cooler has A build in fun size of 120mm enabling higher Aire follow.
Storage smasung (990 EVO Plus 2TB SSD M.2 PCLe 5.0 X2 NVME)	2	Form factor M.2-2280, Interface: M.2 PCLe 5.0*2 Type: SSD	£246.99	Samsung 990 EVO Plus 2TB SSD M.2 PCIe 5.0 X2 NVMe	3D visualisation projects use large storage space, Samsung storage (990 EVO Plus 2TB SSD M.2 PCLe 5.0 X2 NVME) provid high Storage Space and fast performance.
Power supply (MSI MAG A750GL PCIE5 750W)	2	Form factor: ATX, The number of SATA power connections is 8	£79.99	MSI MAG A750GL PCIE5 Black 750W Fully Modular 80+ Gold Certified	This power supply is needed to connect the £D workstation to the power.
Case Fan (Lian Li Uni fan SL-infinity 140mm ARGB PWM Reverse)	2	The minimum noise level = 28.6 dB, Size 140 mm, The minimum air follow is 71.6 CFM	£25.98	Lian Li Uni Fan SL-Infinity 140mm ARGB PWM Reverse	This fan case will maintain the workstation Airflow and remove the temperature generated by the workstation component.

Monitor (Asus ROG Strix XG27ACS 27" 1440p 180Hz IPS Monitor)	4	Size is 27 in Panel type is IPS, refresh rate 180Hz. The number of HDMI is 2.0 port.	£437.98	Asus ROG Strix XG27ACS 27" 1440p 180Hz IPS Monitor	(Asus ROG Strix XG27ACS 27" 1440p 180Hz IPS Monitor) is a high-resolution display with excellence color accuracy which makes it suitable for 3D visualization workstation.
Mouse (Logitech M200 Wireless Mouse with silent clicks)	2	Wireless 2.4GHz. The manufacturer is Logitech.	£17.49	Logitech M220 Wireless Mouse with Silent Clicks	Logitech M200 Wireless Mouse is low price which helps to keep the cost low, and its wireless feature will reduce the cables connection around the workstation.
Keyboard (Logitech K780 Wireless Slim Keyboard)	2	Switch type: Tactile, Layout: ANSI	£69.99	Logitech K780 Wireless Slim Keyboard	Logitech K780 Wireless Slim Keyboard helps to keep workstation organized.
Microphone (FIFINE A6T Microphone)	2	USB connectivity type. Feature: Mute Button RGB lighting. Manufacturer: FIFINE	£51.96	FIFINE A6T USB Microphone	Microphones are needed for communication such as zoom calls and presentations while designing the project.
Total cost (For single workstation)			£3255.32		

Total cost * 2			£6510.64		
Role Budget			£7,500		
Under the budget			£989.36		


Parts
Price History
PC BUILD
Xidax Ready to Ship, Ready to Game
Case
Corsair 3200D RS Black
£59.99
Buy
CPU
AMD Ryzen 9 9950X3D
£499.99
Buy
Motherboard
Asus X870E ProArt CREATOR WIFI AM5 DDR5 ATX
£449.99
Buy
GPU
+ Add
Gigabyte GeForce RTX 5060 WINDFORCE MAX OC 8G
£439.99
Buy
RAM
+ Add
Kingston FURY Beast 64GB (2×32GB) DDR5 5600 CL40 Black
£799.99
Buy
CPU Cooler
Configure
ARCTIC Liquid Freezer III Pro A-RGB Water 360mm 77 CFM Black
£74.99
amazon
Buy

Total Costs Table

Role	No. of workstations	Hardware cost per workstation	Total role cost	Under/over budget
Administrative Assistant	3	Budget: £950 Cost: £740	Budget: £2850 Cost: £ 2220	Under £630
Project Manager	2	Budget: £1300 Cost: £1223	Budget: £2600 Cost: £2446	Under £154
Architect / CAD Designer	5	Budget: £2500 Cost: £2440	Budget: £12500 Cost: £12200	Under £300
3D Visualisation Specialist	2	Budget: 3750 Cost: £3255	Budget: £7500 Cost: £6510	Under £990
Total hardware workstation cost	**12**	-	**Budget: £25450** **Cost: £23376**	**Under £2074**

We acknowledge there is a hefty £2074 in leftover funds, but we believe that the current workstation builds for each role are more than capable to complete their tasks efficiently. We do not see the need in spending unnecessary costs to improve component quality further, when it can be spent elsewhere and have much more value/benefit to the business – like marketing, subscription costs, etc.

SQL Database

When building the database, we ensured it was built with integrity by implementing a unique Role ID, Build ID, and Item ID so the tables are joined and can be queried with ease afterwards. All required fields contain information, and categories use consistent and clear names so someone looking at it for the first time will know what they are looking at.

We will test that the SQL database outputs the correct data when queried, using the SELECT function.

The site used will be https://onecompiler.com/mysql

Queries:

1. List every role

```
SELECT RoleName FROM EmployeeRoles
```

```
+-------------------------------------+
| RoleName                            |
+-------------------------------------+
| Administrative Assistant (Zidan)    |
| Project Manager (N/A)               |
| Architect (Roble)                   |
| 3D Visualisation Specialist (Farouk) |
+-------------------------------------+
```

2. Just the names and budget

```
SELECT RoleName, BudgetPerUnit FROM EmployeeRoles
```

RoleName	BudgetPerUnit
Administrative Assistant (Zidan)	950.0
Project Manager (N/A)	1300.0
Architect (Roble)	2500.0
3D Visualisation Specialist (Farouk)	3750.0

3. The higher-budget roles (over £1000)

```
SELECT RoleName, BudgetPerUnit FROM EmployeeRoles WHERE BudgetPerUnit > 1000
```

RoleName	BudgetPerUnit
Project Manager (N/A)	1300.0
Architect (Roble)	2500.0
3D Visualisation Specialist (Farouk)	3750.0

4. Roles by team size

```
SELECT RoleName, StaffCount FROM EmployeeRoles ORDER BY StaffCount DESC;
```

RoleName	StaffCount
Architect (Roble)	5
Administrative Assistant (Zidan)	3
Project Manager (N/A)	2
3D Visualisation Specialist (Farouk)	2

5. Find one build's items

```
SELECT * FROM BuildItems WHERE BuildID = 1
```

ItemID	BuildID	HardwareCategory	ItemName	Specification	Cost
1	1	CPU	Intel Core i3-12100	4 Cores / 8 Threads, 3.3 GHz, LGA1700 Socket	120.00
2	1	Motherboard	MSI PRO H610M-E DDR4	Micro-ATX, Intel H610, 2x DIMM up to 64GB 3200MHz, PCIe 4.0, M.2 NVMe, HDMI/VGA	60.00
3	1	RAM	CORSAIR Vengeance LPX Black	16GB (2x8GB) DDR4 3200MHz CL16 Dual Channel	145.00
4	1	Storage	Kingston NV3 M.2 2280 - SNV3S/1000G	1TB NVMe SSD Read 6000MB/s, Write 4000MB/s, PCIe 4.0	150.00
5	1	GPU	Integrated Intel UHD Graphics 730	N/A - Integrated with CPU	0.00
6	1	PSU	MSI MAG A550BN	550W 80 PLUS Bronze Power Supply	45.00
7	1	Case/cooling	Fractal Design Core 1100 Black + Intel Laminar RM1 Boxed Cooler	Micro-ATX Mini Tower Case, and the included cooler from the CPU	40.00
8	1	Monitor	PHILIPS 24E1N1200A IPS Monitor - Black	24", 1080p, 120Hz, 1ms, Speakers, HDMI/VGA/DP	75.00
9	1	Keyboard/mouse	Logitech Combo MK270 Combo	Wireless Desktop Keyboard and Mouse USB	25.00
10	1	Webcam	Logitech Brio 100 Full HD Webcam - Graphite	1080p, Auto-Light, Privacy Shutter, Mic	40.00
11	1	Headset	Logitech H390 Stereo MIC USB Black	Noise-Cancelling, USB-A Plug-and-Play	25.00
12	1	Network/cables/adapters	TP-Link Archer T2U Plus	USB 2.0, Dual Band Wi-Fi 5 802.11ac, 5dBi Antenna	15.00
13	1	Role-specific hardware peripheral	N/A	N/A	0.00
14	1	Total hardware workstation cost			740.00
15	1	Role hardware budget			950.00
16	1	Under hardware budget			210.00

6. Count the items in a build

```
SELECT BuildID, COUNT(*) AS ItemCount FROM BuildItems WHERE BuildID = 1
```

```
+---------+-----------+
| BuildID | ItemCount |
+---------+-----------+
|       1 |        16 |
+---------+-----------+
```

7. Total cost of one build

```
SELECT BuildID, Cost AS TotalHardwareCost FROM BuildItems WHERE BuildID = 1 AND HardwareCategory = 'Total hardware workstation cost'
```

```
+---------+-------------------+
| BuildID | TotalHardwareCost |
+---------+-------------------+
|       1 |            740.00 |
+---------+-------------------+
```

Data Integrity

The database reduces inconsistent or invalid data by utilizing:

Primary/foreign keys – primary keys provide unique identification IDs to certain rows, like RoleID, BuildID, ItemID etc. Foreign keys are used to connect tables, for example RoleID is the primary key in EmployeeRoles but also the foreign key in WorkstationBuilds. The RoleID gives a unique ID to each role in the EmployeeRoles table, then this ID is used in WorkstationBuilds to connect the build information to each role.

Relationships between tables – links tables that contain related information using foreign keys. This makes it easier to connect tables to each other without redundancy/repeating information.

Column types – each column holds a suitable data type. For example: columns that hold numbers or IDs are using the INTEGER data type, columns that use text use the TEXT or VARCHAR (if you want to add a character limit) data type, columns that hold decimals use the DECIMAL data type, etc.

Headers - added to each section so someone viewing it clearly knows what they are looking at. This includes information like when a table is created, when data is inserted into the table, who owns each build, what is currently selected to output, etc. These are shown in the green lines that start with '--', meaning it does not affect the code and is purely there for note purposes.

Ethics/Professionalism

It is important to ensure your purchases are responsible, as it stops your business from spending unnecessary funds on something which can have more benefit being spent elsewhere. This can be done by following the recommended practices: in-depth planning, transparent communication with suppliers, purchasing durable but reliable products, build strong relationships with suppliers, avoid last-minute order changes, effective but not aggressive negotiations, etc.

It is also especially important to ensure there is secure handling of business and client information, because it prevents your business from facing legal penalties e.g. not being in compliance with the GDPR/Data Protection Act can lead to heavy financial penalties which can further snowball into reputational damage and cause negative long-lasting effects to the business.

Conclusion

To summarise what we have done: we have identified each role and understood what type of tasks they complete. We then used this information to filter our searches for suitable hardware components and use them in the final build. A component table for each role was made to store specifications, cost, and justification. We compiled all this information into an SQL database, tested it

worked by using some SELECT queries, explained how the database kept its integrity, and concluded the report with an explanation about responsible purchasing and the importance of secure data handling.